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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H3 CHEMISTRY

9813/01

Paper 1

24 September 2021

INSERT

2 hours 30 minutes

INSTRUCTIONS

This insert contains information for Question 1. **Do not write your answers on the insert.**

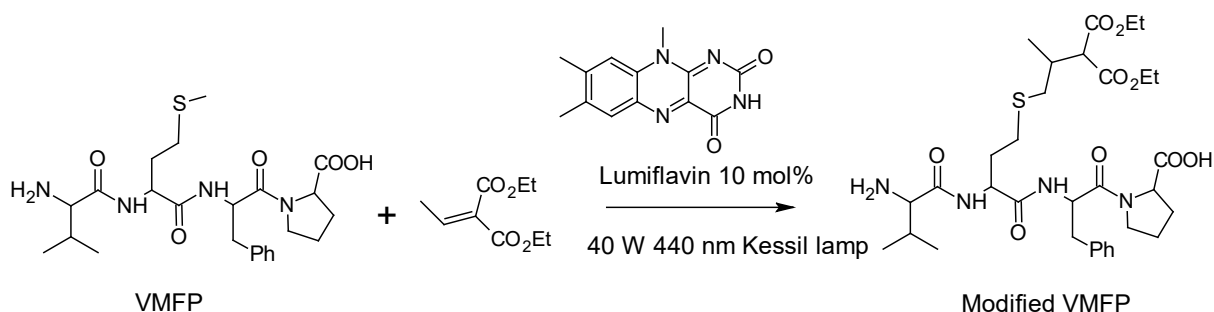
Information for Question 1

Photoredox catalysis and electrochemistry are important synthetic methods that expand the toolbox of the organic chemist. One important goal of synthetic organic chemistry is the activation of a C–H bond. While transition metal based photocatalysts remain widely used, there has been growing interest in the development of organic photoredox catalysts.

Abstract 1 summarises a published research paper on a photoredox method and Abstract 2 summarises another paper showcasing an electrochemical method of C–H bond activation.

Abstract 1 Photoredox method (*J. Am. Chem. Soc.* 2020, 142, 21260-21266)

Bioconjugation technologies have revolutionised the practice of biology and medicine by allowing access to novel biomolecular scaffolds. New methods for residue selective bioconjugation are highly sought after to expand the toolbox for a variety of bioconjugation applications. MacMillan et al demonstrated the use of lumiflavin as an organic photoredox catalyst to perform bioconjugation through the selective alkylation of the amino acid methionine.



Bioconjugation of VMFP

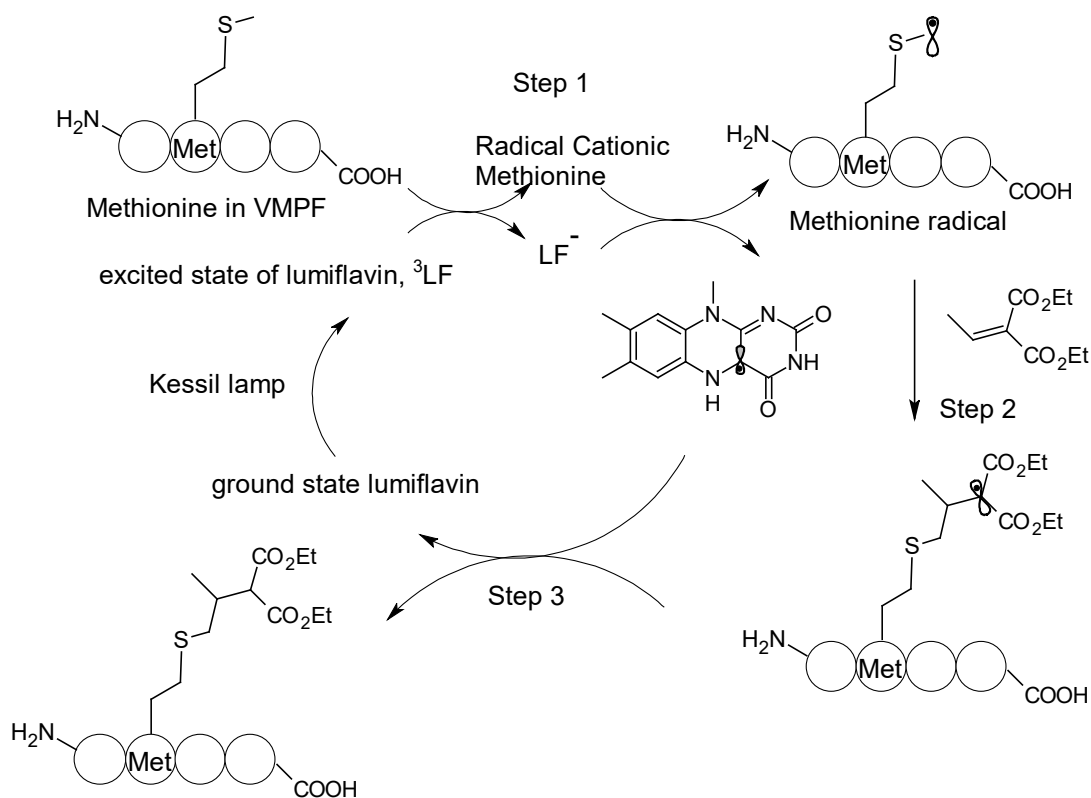
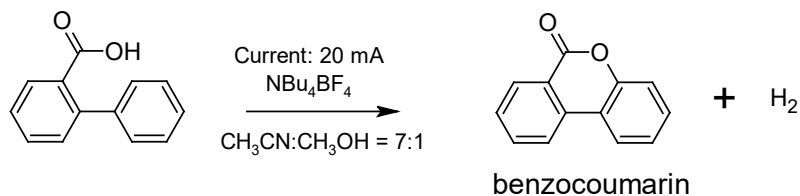


Fig. 1.1 Lumiflavin Catalytic Cycle

Abstract 2 Electrochemical method (*Org. Lett.* 2018, 20, 252-255)

Lactones (cyclic esters) are ubiquitous in molecules of biological and pharmaceutical interest. Classic Fischer esterification or acylation methods require the use of a hydroxyl nucleophile. Zeng et al has developed an electrochemical lactonisation method which involves the activation of a C–H bond. It does not require the use of an external oxidant or catalyst and is a highly atom economical reaction.



The proposed mechanism for the formation of benzocoumarin is outlined in Fig. 1.2.

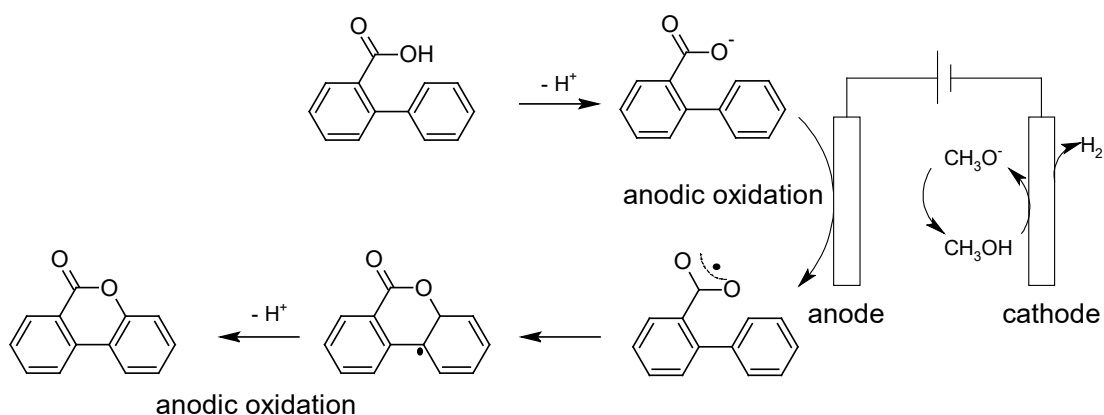


Fig. 1.2

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© Junyong Kim, Beryl X. Li, Richard Y.-C. Huang, Jennifer X. Qiao, William R. Ewing, and David W. C. MacMillan
J. Am. Chem. Soc. 2020, 142, 21260-21266

© Sheng Zhang, Lijun Li, Huiqiao Wang, Qian Li, Wenmin Liu, Kun Xu, and Chengchu Zeng
Org. Lett. 2018, 20, 252-255

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Org. Lett. 2015, 17, 18, 4550–4553